

SoS, DSD and SECO: Relations among these Areas

By Claudia Werner

COPPE/UFRJ

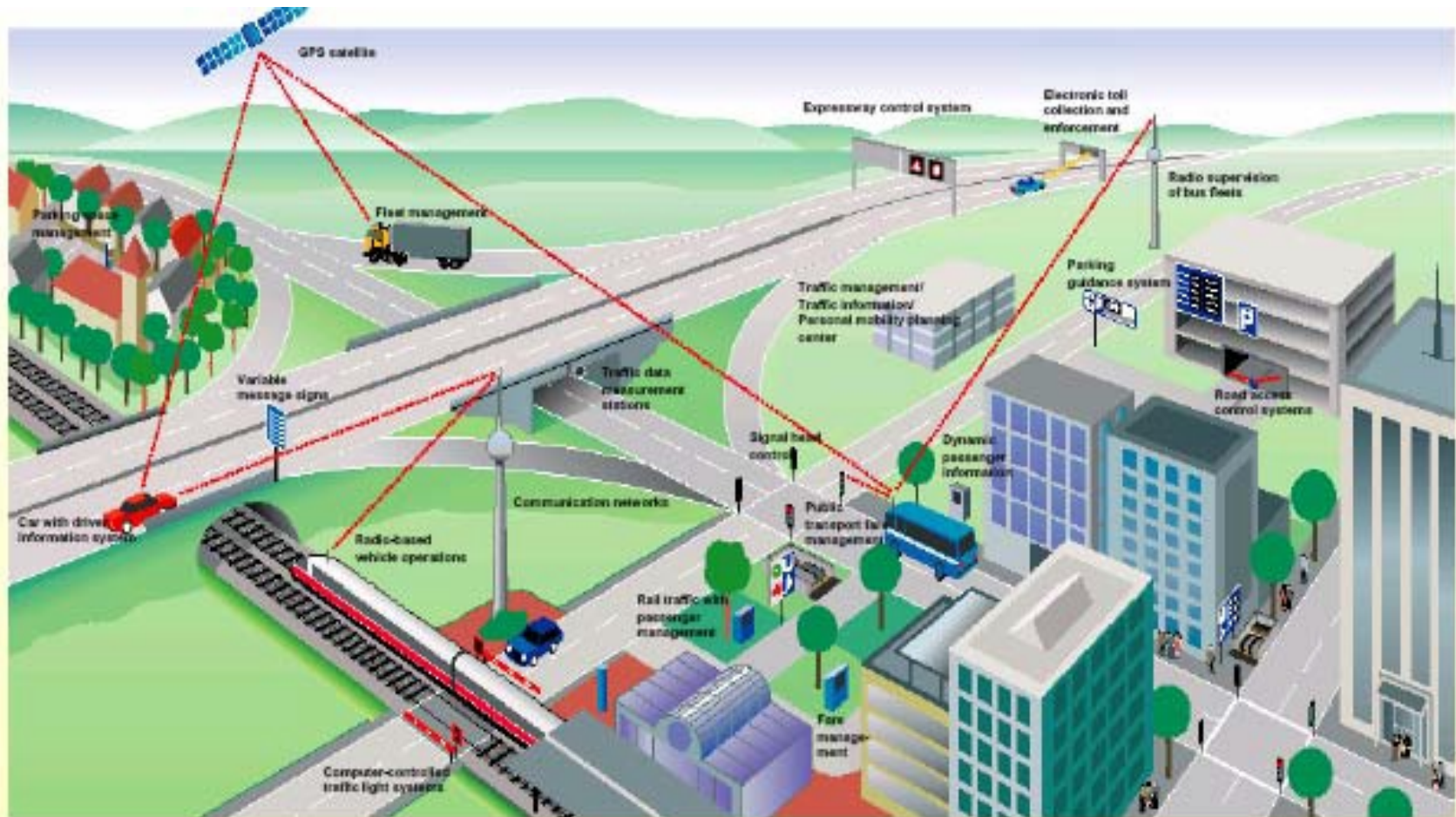


ICSE
4th International Workshop on
Software Engineering for Systems-of-Systems
Austin, Texas, USA
Monday, May 16, 2016

Workshop on DSD, SECO and SoS



SoS – Systems of Systems



(European Commission)

SoS – Systems of Systems

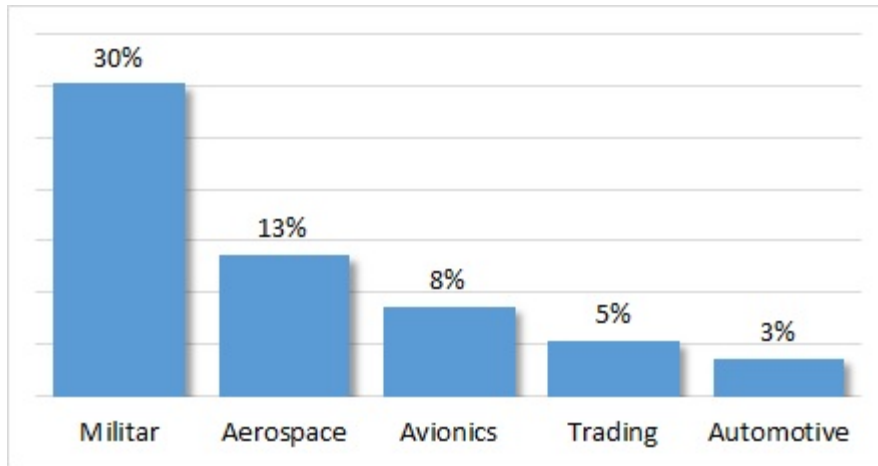
- General characteristics:
 - Constituent Systems (CSs) operational independence
 - CSs managerial independence
 - CSs geographical distribution
 - SoS evolutionary development
 - SoS emergent behaviors

SoS – Systems of Systems

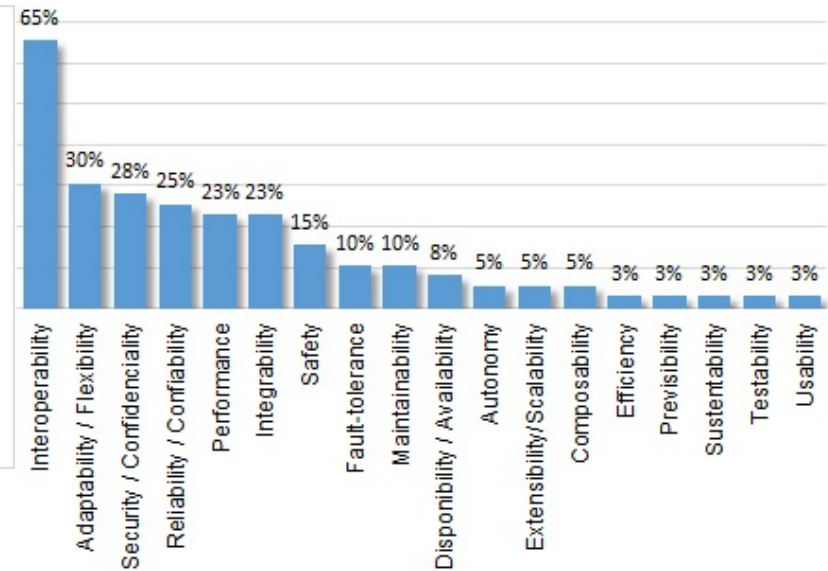
- Software-intensive System-of-systems refers to a set of organizational, managerial independent **constituent systems**, mainly coordinated by **software systems**, and collaborating among them to comply with a comprehensive, new **mission** in a **dynamic** context with regard to the **environment**, the **constituents**, and even the **mission**.

[Nakagawa, 2014]

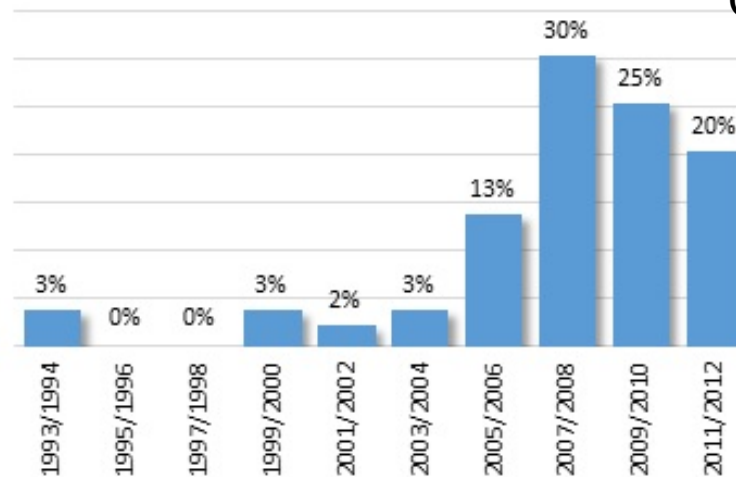
SoS – Systems of Systems



Application Domains



Quality Attributes



Publication Distribution through Years



SoS – Systems of Systems

- Some challenges:
 - Adequately deal with SoS software architectures
 - Investigate how to better develop SoS for diverse domains
 - Propose solutions to different types of SoS

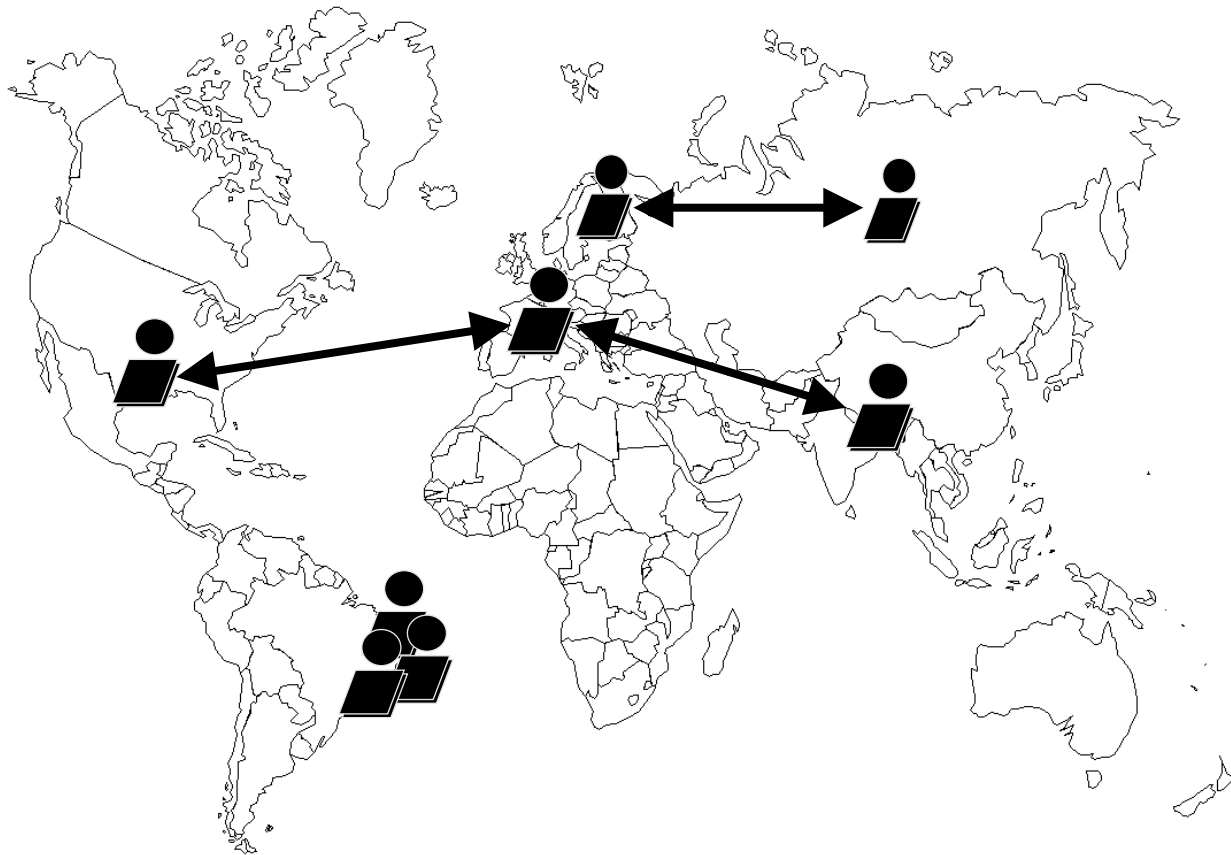
What are the fundamental challenges we need to address in the Software Engineering Community?



By Elisa Yumi
Nakagawa

ICMC/USP

DSD – Distributed Software Development



DSD – Distributed Software Development

- It aims to help companies to:
 - Be closer to local markets
 - Find local talents
 - Speed up development cycle
 - Reduce developments costs

**Do we have evidence these expected
benefits are being achieved?**

DSD – Distributed Software Development

- Challenges:
 - Communication
 - Coordination
 - Collaboration
 - Delays
 - Language barriers
 - Etc...

**By Sabrina
Marczak**

PUCRS



DSD – Distributed Software Development

- Research challenges:
 - Local cultures and perspectives
 - Language barriers
 - Access to remote sites
 - Budget to remote on-site studies
 - Etc...

By Sabrina
Marczak

PUCRS

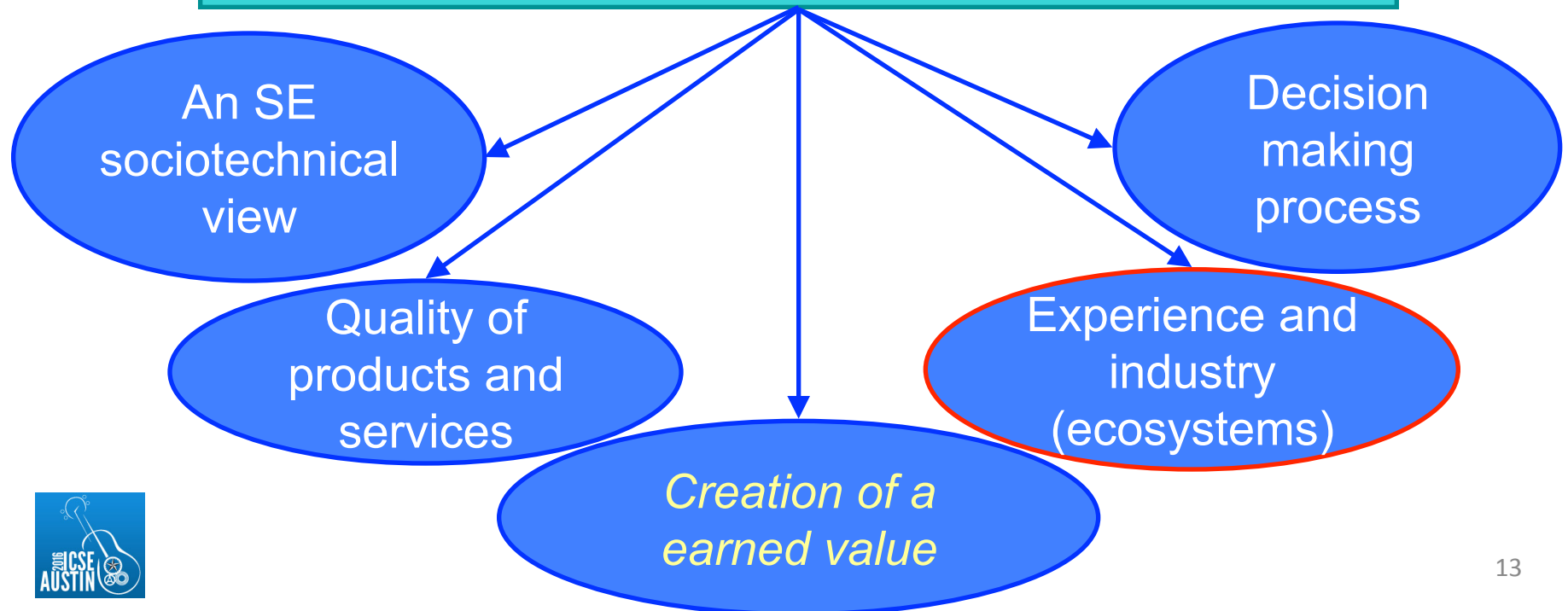


SECO – Software Ecosystems

- Software Engineering Challenges

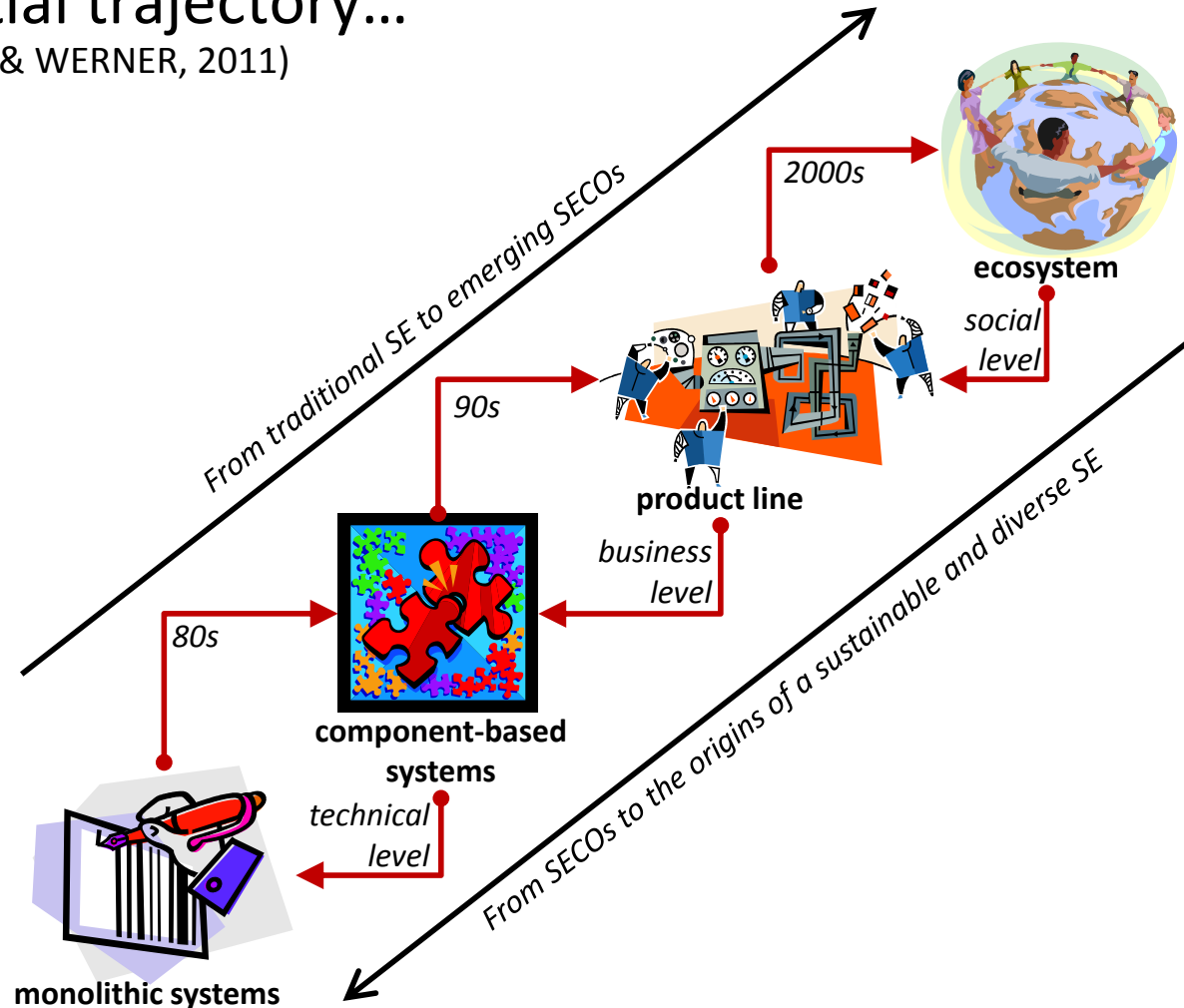
(BOEHM & SULLIVAN, 2000, BIFFL et al., 2006, CUKIERMAN et al., 2007, WERNER et al., 2009)

***Incorporation of economic and social issues in
Software Engineering (SE)***



SECO – Software Ecosystems

- An initial trajectory...
(SANTOS & WERNER, 2011)



SECO – Software Ecosystems

- A software ecosystem is a set of actors functioning as a unit and interacting in a shared market for software and services, based on a common technical platform or market.

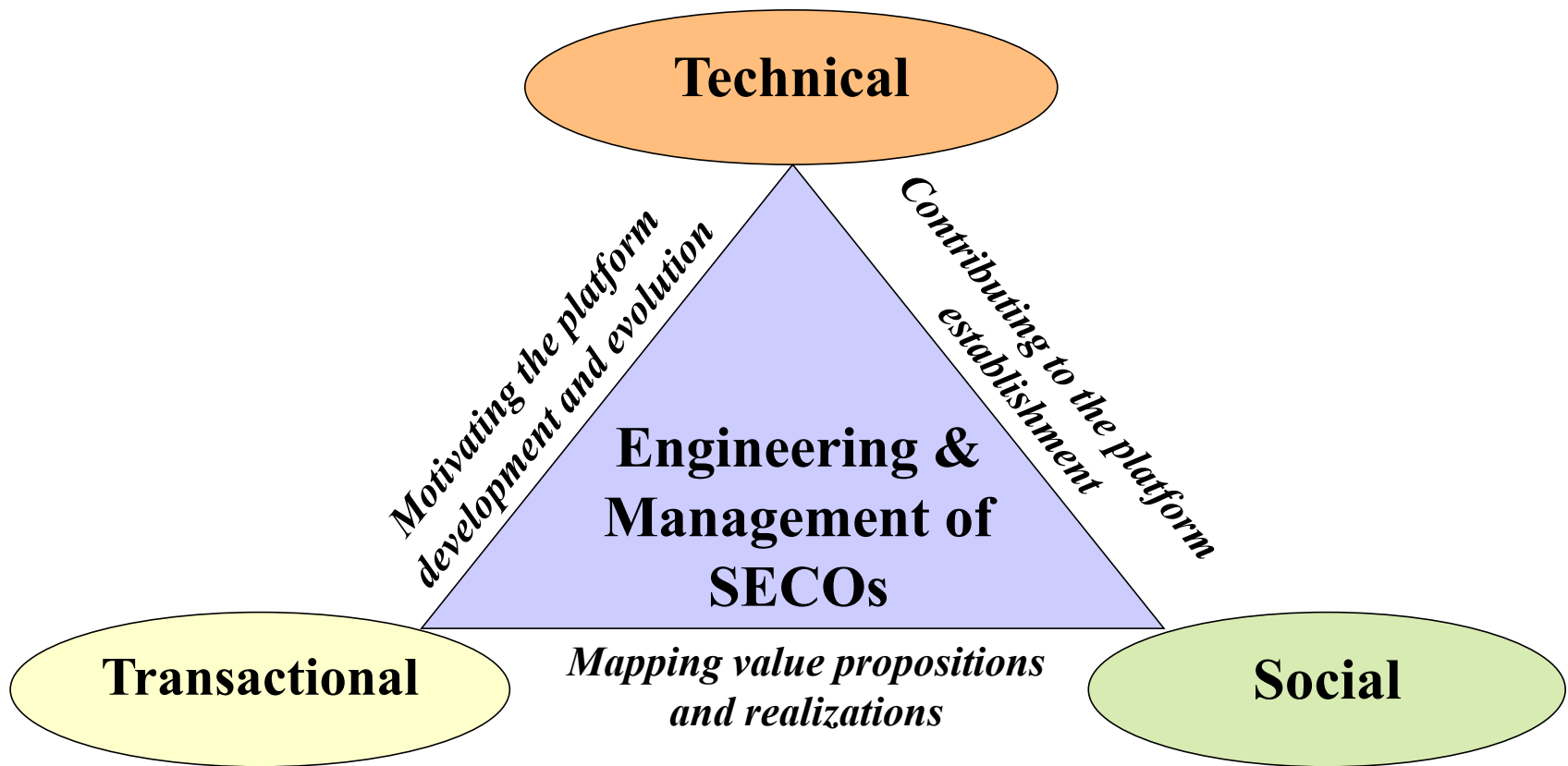
(JANSEN et al., 2009)

- Software ecosystem consists of a set of software solutions that support and automate activities and transactions of actors who are associated with a social or business ecosystem.

(BOSCH, 2009)

SECO – Software Ecosystems

- Dimensions



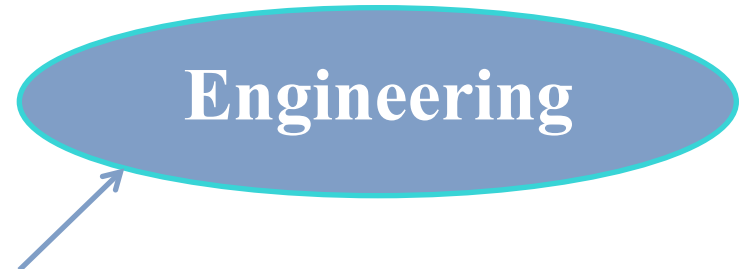
SECO – Software Ecosystems

- Challenges

Open Source Ecosystems

Governance

Analysis



Openness

Quality

Software Architecture



PANEL

Impacts of SECO in the Quality of DSD and SoS

SECO and SoS

(Santos et al., 2014b) (Jeronimo Jr and Werner, 2015)

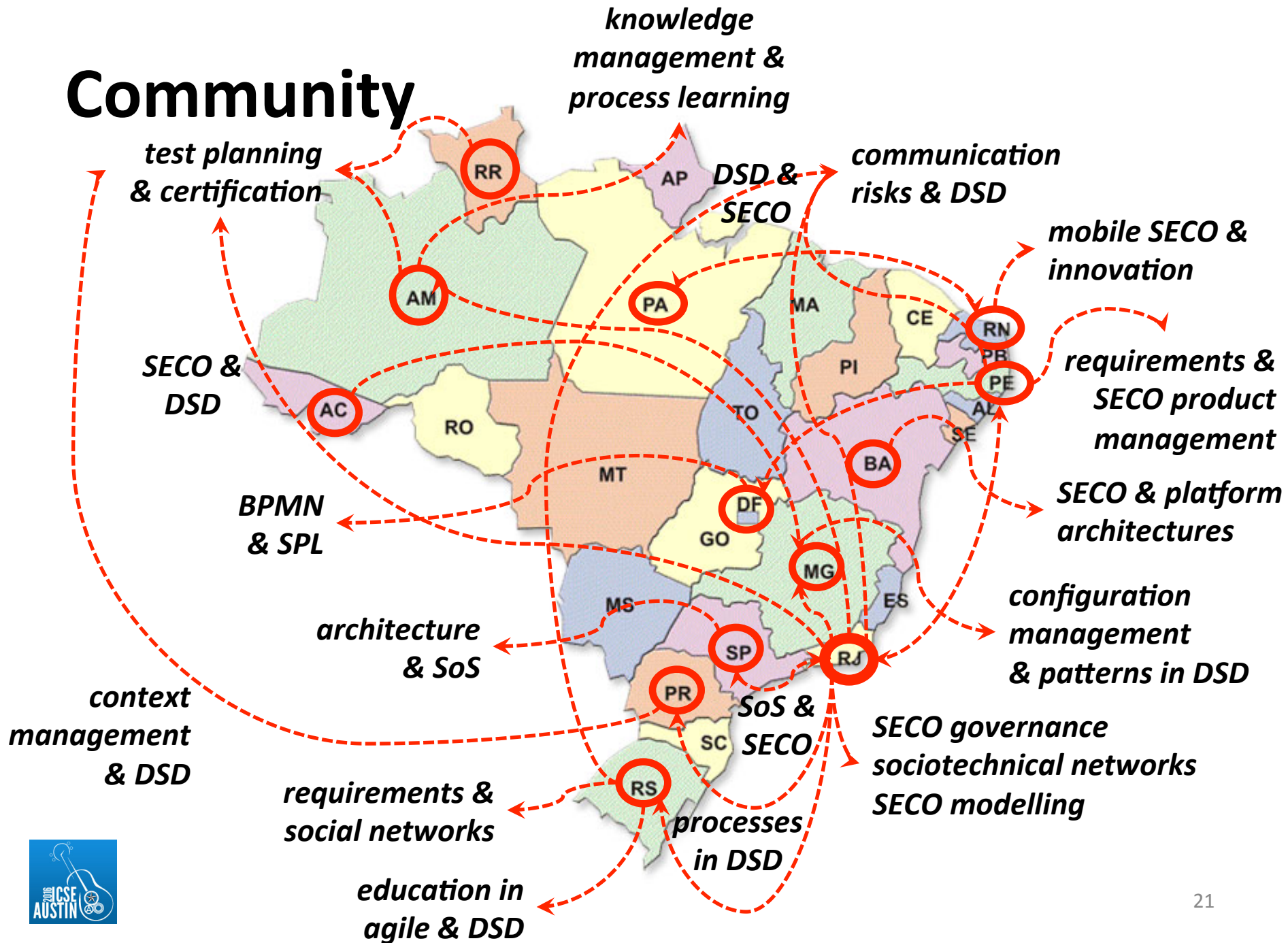
- Big, complex, heterogeneous and distributed systems (independent components)
- Dynamic evolution and reuse
- Social and Business aspects, besides technical
- Coordination, Communication and Collaboration aspects
- SoS platform (stability, security, interoperability and reliability)

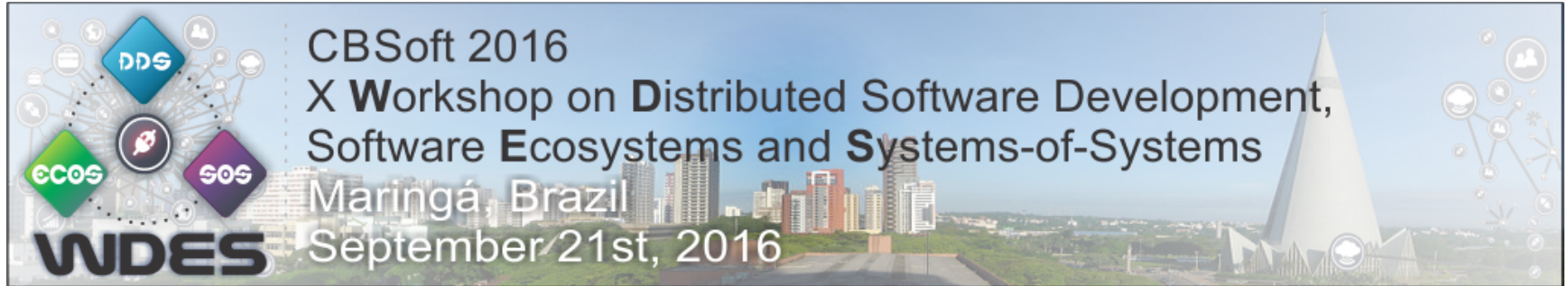
SECO and DSD

(Santos et al., 2014a)

- Globalized, large-scale and long-term platform Development
- Process alignment
- *Outsourcing* governance
- Hybrid business model
- Sustainability and Diversity
- Partner selection (collaborators)
- Architecture opening

Community





CPF

Submission deadline: June 10th

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